

No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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1 Abstract

Stackelberg Security Games (SSGs) model the strategic allocation of limited defender resources against attackers whose preferences are often uncertain. Balcan et al. [2] designed no-regret online learning algorithms for the *repeated* variant of SSGs against an adversarial sequence of attackers drawn from a finite, *known* set K , and proved a matching linear-regret lower bound when K is unbounded. A natural real-world middle ground is left open: the defender does not know K a priori, but has a bound $K_{\max}$ on its size. We study this setting in the full-information feedback model. We propose GROW-CATALOGUE, an algorithm that incrementally learns new attacker types as they arrive, re-partitions the coverage polytope, and runs an anytime multiplicative-weights sub-routine on the resulting extreme-point set. Our central technical contribution is a full proof that GROW-CATALOGUE achieves expected regret

$$O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right)$$

against any oblivious adversarial sequence of length T . The rate matches that of Balcan et al. [2] with k replaced by $K_{\max}$ and is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$. Experiments on synthetic SSG instances confirm the predicted $\sqrt{T}$ scaling, and the empirical regret stays comfortably below the analytic $C\sqrt{T}$ envelope. Our implementation and all experiment scripts are at github.com/aditisingh1053/Security_Games.

2 Introduction

In a Stackelberg Security Game (SSG), a defender allocates a scarce set of resources to protect n potential targets against a strategic attacker who, observing the defender's randomized deployment, chooses the best target to attack [16]. Deployed systems built around this model include the US Coast Guard's port protection patrols, the Federal Air Marshals assignment system, the LAX airport patrol schedule, and anti-poaching patrols in several African national parks [16, 17]. In every one of these applications the defender must commit to its deployment before the attack happens, and the attacker's payoff function encodes its preferences over targets.

The *classical* SSG analysis assumes that the attacker's payoff function is fully known to the defender. The optimal commitment can then be computed by a linear program [e.g., 15]. In practice, however, the defender faces considerable uncertainty about the attacker. There may be several distinct threat actors (e.g., different poacher groups, different types of smugglers), and their utilities are only imperfectly known. A principled response, pioneered by Balcan et al. [2], is to drop the single-type assumption entirely and instead ask for

a *no-regret* online algorithm against an adversarially chosen sequence of attacker types, drawn from a known finite set K with $|K| = k$. Under full-information feedback, they design an algorithm whose expected cumulative regret is bounded by $O(\sqrt{Tn^2k \log(nk)})$, which is sublinear in the horizon T , and they prove a matching linear-regret lower bound when $|K|$ is unrestricted [2, Theorem 7.1].

The assumption that the set K itself is known to the defender in advance is, however, a strong one. In realistic deployments the defender typically does *not* have a complete list of attacker types at the start. What it may have is an *upper bound* $K_{\max}$ on the number of distinct types it expects to encounter (for instance, from threat intelligence or from physical constraints of the domain). The question that motivates this project is therefore:

Can the defender achieve sublinear regret when K is unknown a priori, subject to $|K| \leq K_{\max}$?

This question lies squarely in the gap left open between the *known- K* upper bound and the *unrestricted- K* lower bound of Balcan et al. [2]. It is not answered by any of the existing work, and resolving it is the main goal of the present report.

2.1 Contributions

Our contributions are the following.

- **Problem formalization (Section 4).** We formalize the unknown- K variant of the repeated Stackelberg security game, identifying the minimal modification of the model of Balcan et al. [2] that captures it.
- **Algorithm (Section 5).** We propose GROW-CATALOGUE, an online algorithm that at each round maintains a *catalogue* $C_t \subseteq K$ of attacker types observed so far, recomputes the set $\mathcal{E}(C_t; \epsilon)$ of ϵ -approximate extreme points of the partition of $\mathcal{P}$ induced by C_t , and runs an anytime multiplicative-weights sub-routine over $\mathcal{E}(C_t)$. When a new type is observed, the catalogue is updated and the sub-routine is restarted on the enlarged expert set.
- **Main theorem and a complete, detailed proof (Sections 5.2 and 6).** The technical core of this report is a self-contained proof that GROW-CATALOGUE achieves expected regret at most $O(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)})$ against any oblivious adversarial sequence of length T under full-information feedback. The rate matches Theorem 5.1 of Balcan et al. [2] with k replaced by $K_{\max}$; for constant $K_{\max}$ it is $\tilde{O}(n\sqrt{T})$.
- **Regime analysis (Corollaries 7 and 8).** We characterize the parameter regime in which the bound is meaningful ($K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$), and we prove an instance-adaptive refinement in which the leading factor depends on

the number K' of types *actually* observed rather than on $K_{\max}$.

- **Experiments (Section 7).** We implement GROW-CATALOGUE and two reference baselines, and evaluate them on synthetic SSG instances. The empirical regret scales as predicted by the theorem, staying comfortably below the analytic $C\sqrt{T}$ envelope.

2.2 Related Work

Classical SSGs and attacker uncertainty. The optimal commitment for a known attacker can be computed by an LP [5]. Uncertainty about the attacker has been handled by *robust optimization*, which plans against worst-case payoffs [15], and by *Bayesian SSGs*, which require a prior over attacker types [10, 13]. Both approaches rest on priors or constraints that are often hard to validate. Letchford et al. [11] give a query-efficient learning algorithm for a *single* fixed but unknown type, and Blum et al. [3] extend this to a setting that adapts round-by-round to one fixed type.

Online learning against adversarial attacker sequences. Balcan et al. [2] pivot from the single-type paradigm by allowing the sequence of attackers to be adversarially chosen from a known finite set K . For full information they give a polynomial-weights algorithm with regret $O(\sqrt{Tn^2k \log(nk)})$ that partitions the coverage polytope into best-response regions and runs experts over the ϵ -approximate extreme points; for bandit feedback they use a barycentric spanner construction to give an $O(T^{2/3})$ bound. Crucially, they also prove a linear lower bound when $|K|$ is unrestricted, which forces any non-trivial extension to restrict the attacker type set somehow. We take the weakest restriction: a known cardinality bound $K_{\max}$.

Online learning with changing expert sets. The subroutine inside GROW-CATALOGUE is an anytime variant of Polynomial Weights / Hedge [4, 6, 12]. We restart this subroutine each time the catalogue grows; simpler algorithms for tracking a growing set of experts exist [14], and for tracking a shrinking set [7], but both would only shave lower-order logarithmic factors.

Other recent directions. Haghtalab et al. [8] analyze no-regret learning against strategic (non-myopic) attackers in a single-type query model. Harris et al. [9] extend to a contextual Stackelberg setting in which both players observe side information. Neither addresses the *unknown K with bounded cardinality* question that we pose here.

3 Model

We adopt the repeated-SSG notation of Balcan et al. [2, Sec. 2]; our notation is compatible with theirs.

3.1 Game primitives

The game has a finite horizon $T \in \mathbb{N}$ and n targets $N = \{1, \dots, n\}$. The defender has R resources and a collection $\mathcal{D} \subseteq 2^N$ of *schedules*, where each schedule is a subset of

targets that a single resource can defend simultaneously. An *assignment function* $A : R \rightarrow 2^{\mathcal{D}}$ records the schedules that each resource may be assigned to. A *pure strategy* is any valid resource-to-schedule assignment; let m denote the number of pure strategies. A *mixed strategy* is a distribution $\mathbf{s} \in \Delta_m$ over pure strategies, and its induced *coverage probability vector* is $\mathbf{p} = M\mathbf{s} \in [0, 1]^n$, where $M \in \{0, 1\}^{n \times m}$ has $M_{ij} = 1$ iff the j -th pure strategy covers target i . Write $\mathcal{P}$ for the set of valid coverage probability vectors; Every coverage vector $\mathbf{p}$ is realizable by some mixed strategy, and conversely every mixed strategy induces a unique $\mathbf{p} \in \mathcal{P}$.

Defender payoffs. For each target i , the defender has a payoff $u_d^c(i) \in [-1, 1]$ if i is attacked *and* covered, and a payoff $u_d^u(i) \in [-1, 1]$ if i is attacked *and* uncovered. The defender's expected utility when target i is attacked under coverage $\mathbf{p}$ is therefore

$$U_d(i, \mathbf{p}) = u_d^c(i) p_i + u_d^u(i) (1 - p_i), \quad (1)$$

which is *linear* in $\mathbf{p}$ for each fixed target i .

3.2 Attacker types and best response

An attacker *type* α is a pair $(u_\alpha^c, u_\alpha^u) \in [-1, 1]^n \times [-1, 1]^n$ of per-target utilities following the same “covered/uncovered” convention as the defender. Its expected utility for attacking target i under coverage $\mathbf{p}$ is $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i) p_i + u_\alpha^u(i) (1 - p_i)$. Observing $\mathbf{p}$, type α *best-responds*:

$$b_\alpha(\mathbf{p}) = \arg \max_{i \in N} U_\alpha(i, \mathbf{p}), \quad (2)$$

with ties broken arbitrarily but consistently. Note that b_α is piecewise constant in $\mathbf{p}$, which makes the defender's realized payoff $U_d(b_\alpha(\mathbf{p}), \mathbf{p})$ a discontinuous function of $\mathbf{p}$. This non-continuity is the central technical difficulty that distinguishes SSGs from vanilla online linear optimization, and it is what drives the polytope-based machinery of Balcan et al. [2].

3.3 Repeated game and regret

There is a (possibly unknown) set K of attacker types. An *oblivious* adversary chooses a sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ before the game starts; the sequence is fixed but unknown to the defender. At each round $t \in \{1, \dots, T\}$ the defender commits $\mathbf{p}_t \in \mathcal{P}$, attacker a_t plays $b_{a_t}(\mathbf{p}_t)$, and the defender receives the random payoff $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$. We assume *full-information feedback*: at the end of round t the defender observes the identity of a_t , and the first time any type α is observed, its utility functions are revealed as well.

The defender's benchmark is the best fixed coverage vector in hindsight:

$$\mathbf{p}^* = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}), \mathbf{p}). \quad (3)$$

The expected regret of an online algorithm that plays $\mathbf{p}_t$ at round t is

$$\text{Regret}(T) = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right], \quad (4)$$

where the expectation is over the algorithm's internal randomization.

3.4 Preliminaries: polytope partition and $\mathcal{E}(C; \epsilon)$

The following definitions, taken from Balcan et al. [2, Sec. 4], are essential. For any catalogue $C \subseteq K$, a type $\alpha \in C$, and a target $i \in N$, the region in which α 's best response is i is

$$\mathcal{P}_i^\alpha(C) = \{\mathbf{p} \in \mathcal{P} : b_\alpha(\mathbf{p}) = i\}, \quad (5)$$

which is the intersection of $\mathcal{P}$ with the $n-1$ linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$ for $j \neq i$; hence a convex polytope. For a response profile $\sigma : C \rightarrow N$ we write

$$\mathcal{P}_\sigma(C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(C). \quad (6)$$

The non-empty sets $\mathcal{P}_\sigma(C)$ form a partition of $\mathcal{P}$ into convex regions in which the best response of every type in C is held fixed; within any such region, the defender's payoff under a fixed attacker type is a linear function of $\mathbf{p}$.

We will need three results from Balcan et al. [2], restated below in our notation and cited explicitly at every use. Lemmas 1 and 2 describe the approximate extreme-point set; Proposition 3 is the standard Polynomial-Weights online learning guarantee, adapted to the present loss range.

Lemma 1 (2, Lemma 4.3). *For every $\epsilon > 0$ and every catalogue $C \subseteq K$, there is a finite set $\mathcal{E}(C; \epsilon) \subseteq \mathcal{P}$ of points, called the ϵ -approximate extreme points of the partition, such that for any sequence $\mathbf{a}' = a'_1, \dots, a'_{T'}$ with $a'_t \in C$ for all t ,*

$$\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T'.$$

Lemma 2 (2, Lemma 4.4). *The set $\mathcal{E}(C; \epsilon)$ of Lemma 1 can be chosen so that $|\mathcal{E}(C; \epsilon)| \in O((2^n + |C|n^2)^n \cdot n^{|C|})$.*

Proposition 3 (2, Proposition 1). *For any fixed expert set $\mathcal{M}$ and any sequence of losses in $[-\kappa, \kappa]$, there is an online learning algorithm (an anytime variant of Polynomial Weights / Hedge; see 4, Sec. 2) whose expected regret after T' rounds is at most $O(\sqrt{T' \kappa \log |\mathcal{M}|})$, without requiring T' to be known in advance.*

4 The Unknown- K Setting

We now state the problem that our algorithm and theorem address.

Problem 4 (Unknown- K repeated SSG). A repeated SSG as in Section 3 is played, with two modifications.

1. The set K of attacker types is not revealed to the defender at the start; however, the defender is given a constant $K_{\max} \geq 1$ with the promise $|K| \leq K_{\max}$.
2. Feedback is full information: after round t the defender observes a_t , and the first time any type α appears, its utility vectors (u_α^c, u_α^u) are revealed.

The benchmark $\mathbf{p}^*$ and regret $\text{Regret}(T)$ are as in Equations (3) and (4).

Algorithm 1 GROW-CATALOGUE (full information).

Require: horizon T , bound $K_{\max}$, game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

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1:  $C \leftarrow \emptyset$ ;  $\mathcal{E} \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/\sqrt{T}$ ;  $t_{\text{loc}} \leftarrow 0$ 
2: for  $t = 1, \dots, T$  do
3:   if  $\mathcal{E} = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     sample  $\mathbf{p}_t \sim q_t$ , where  $q_t$  is the current PW distribution
       over  $\mathcal{E}$ 
7:   end if
8:   observe  $a_t$ 
9:   if  $a_t \notin C$  then ▷ discovery round: new attacker type
10:     $C \leftarrow C \cup \{a_t\}$ ; record  $(u_{a_t}^c, u_{a_t}^u)$ 
11:     $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$  ▷ recompute extreme points
12:    reset PW on  $\mathcal{E}$ ;  $t_{\text{loc}} \leftarrow 0$ 
13:   else ▷ routine round
14:     $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$ 
15:    compute  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$  for  $\mathbf{p} \in \mathcal{E}$ 
16:     $\eta \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$ 
17:     $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} \cdot e^{-\eta \ell_t(\mathbf{p})}$  for all  $\mathbf{p} \in \mathcal{E}$ 
18:   end if
19: end for
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Remark 5 (Necessity of the bound $K_{\max}$). Balcan et al. [2, Theorem 7.1] exhibit, for every T , a game with $|K| < 2^{T+1}$ on which every online algorithm has expected regret at least $T/2$. Hence some finite bound on $|K|$ is *necessary* for sublinear regret; Problem 4 takes the weakest such bound, namely a known constant $K_{\max}$.

5 Algorithm and Main Result

5.1 The Grow-Catalogue algorithm

We describe GROW-CATALOGUE at a high level and then give the pseudocode in Algorithm 1. At any round t the defender holds a catalogue $C_t \subseteq K$ of all types seen so far. Conceptually, the defender runs a small instance of the *known- K* algorithm of Balcan et al. [2, Sec. 5] on the catalogue C_t , pretending that C_t is the full attacker set. Concretely:

- The algorithm maintains the ϵ -approximate extreme-point set $\mathcal{E}_t = \mathcal{E}(C_t; \epsilon)$ of Lemma 1. We set $\epsilon = 1/\sqrt{T}$; this balances the $2\epsilon T$ approximation term in Lemma 1 against the $\sqrt{T \log N_{\max}}$ regret term of Proposition 3.
- An anytime Polynomial-Weights sub-routine of Proposition 3 runs over $\mathcal{E}_t$, updating via a time-varying learning rate $\eta_{t_{\text{loc}}} = \sqrt{\log |\mathcal{E}_t| / t_{\text{loc}}}$ where t_{loc} is a local counter reset at every catalogue change. This is the standard anytime schedule; see Cesa-Bianchi and Lugosi [4, Sec. 2].
- When a new attacker type α appears, the catalogue grows to $C_t \cup \{\alpha\}$, the approximate extreme-point set is recomputed, and the sub-routine is restarted on the enlarged expert set. The single round of first contact with α contributes a bounded additive amount to the regret that we will absorb.

5.2 Main Result

Theorem 6 (Main). *Let the defender play Algorithm 1 in the unknown- K full-information setting of Problem 4. For any oblivious adversarial sequence $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$, the*

expected regret satisfies

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right). \quad (7)$$

In particular, if $K_{\max} = O(1)$ then $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(n\sqrt{T})$, matching the known- K bound of Balcan et al. [2, Theorem 5.1] up to logarithmic factors.

We prove Theorem 6 in detail in Section 6. Two useful corollaries of the proof (the regime of applicability and an instance-adaptive refinement) are stated after the proof in Section 6.1.

6 Proof of Theorem 6

The proof closely tracks the structure of the known- K proof of Balcan et al. [2, Sec. 5] but has two non-trivial deviations: (i) the expert set is no longer static but grows over time, and (ii) the discovery rounds introduce an extra bounded additive term. We account for both the deviations in our proof.

Proof:- Fix an oblivious adversarial sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ with $|K| \leq K_{\max}$. Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ denote the number of distinct types that actually appear in $\mathbf{a}$. The proof proceeds in ten steps.

Step 1 (Epoch decomposition). Let $\tau_1 < \tau_2 < \dots < \tau_{K'}$ be the round indices at which each new attacker type is first observed, i.e., for each $i \in \{1, \dots, K'\}$, round τ_i is the smallest t such that $a_t \notin \{a_1, \dots, a_{t-1}\}$. We define the *epochs*

$$\text{Ep}_i = \{\tau_i + 1, \tau_i + 2, \dots, \tau_{i+1} - 1\}, \quad i = 1, \dots, K', \quad (8)$$

with the convention $\tau_{K'+1} := T + 1$, and write $L_i = |\text{Ep}_i|$. Let $\mathcal{D} = \{\tau_1, \dots, \tau_{K'}\}$ be the set of *discovery rounds*. By construction,

$$[1, T] = \mathcal{D} \sqcup \bigcup_{i=1}^{K'} \text{Ep}_i,$$

and $|\mathcal{D}| + \sum_{i=1}^{K'} L_i = T$ with $|\mathcal{D}| = K' \leq K_{\max}$. The epoch decomposition is the critical structural property of the proof: it re-interprets the online problem with a growing expert set as a finite sequence of online problems with *fixed* expert sets, so that the known- K analysis of Balcan et al. [2] can be invoked inside each epoch.

Step 2 (Catalogue is constant inside an epoch). Immediately after round τ_i , the catalogue maintained by Algorithm 1 is $C_i = \{a_{\tau_1}, \dots, a_{\tau_i}\}$, and it is not updated until the next discovery round τ_{i+1} . Hence $C_t = C_i$ for every $t \in \text{Ep}_i$. Moreover, by the defining property of τ_{i+1} , no new type appears in Ep_i , so

$$a_t \in C_i \quad \text{for every } t \in \text{Ep}_i. \quad (9)$$

Denote by $\mathcal{E}_i := \mathcal{E}(C_i; \epsilon)$ the extreme-point set used by the PW sub-routine during Ep_i ; Equation (9) is exactly the hypothesis needed to apply Lemma 1 to C_i and the sub-sequence inside Ep_i .

Step 3 (Within-epoch regret). During Ep_i the algorithm runs a fresh instance of the anytime Polynomial-Weights sub-routine on the fixed expert set $\mathcal{E}_i$ for L_i rounds. The losses are $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so the range parameter is $\kappa = 1$. Proposition 3 therefore yields an absolute constant $c_1 > 0$ such that

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & - \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (10)$$

Step 4 (Within-epoch approximation via Lemma 1).

We now wish to compare the max over $\mathcal{E}_i$ during Ep_i with the *global* benchmark $\mathbf{p}^*$ (the best fixed strategy over the full horizon $[1, T]$). Lemma 1, however, is a statement of the form “ $\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)}(\cdot) \geq \max_{\mathbf{p} \in \mathcal{P}}(\cdot) - 2\epsilon T'$ ” and, strictly speaking, does *not* directly involve $\mathbf{p}^*$: it compares the max over the finite set $\mathcal{E}(C; \epsilon)$ against the unrestricted max over $\mathcal{P}$, both on the *same* attacker sequence. We therefore proceed in two sub-steps and make the passage to $\mathbf{p}^*$ explicit.

Sub-step 4a (apply Lemma 1 to the epoch sub-sequence). By Equation (9), the sub-sequence $(a_t)_{t \in \text{Ep}_i}$ has length L_i and uses only types in C_i , which is exactly the hypothesis of Lemma 1. Applying the lemma with catalogue C_i and sequence length $T' = L_i$ gives

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \underbrace{\max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p})}_{=: \text{OPT}_i} - 2\epsilon L_i. \end{aligned} \quad (11)$$

Here OPT_i is the best cumulative utility achievable on Ep_i by *any* coverage vector in $\mathcal{P}$; crucially, this is the epoch- i optimum, which in general *differs* from the value of $\mathbf{p}^*$. This is the subtle point that distinguishes our setting from the known- K proof of Balcan et al. [2, Sec. 5]: in their analysis, Lemma 1 is applied once on the full horizon, so $\text{OPT} = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*)$ by definition of $\mathbf{p}^*$ and no separate passage is needed. In our per-epoch application, OPT_i is an epoch-local quantity, potentially larger than $\mathbf{p}^*$'s value on the epoch.

Sub-step 4b (pass from OPT_i to $\mathbf{p}^$).* Since $\mathbf{p}^* \in \mathcal{P}$, the coverage vector $\mathbf{p}^*$ is a *feasible* competitor in the maximization defining OPT_i . Therefore

$$\text{OPT}_i \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (12)$$

Inequality Equation (12) is *not* a consequence of Lemma 1 but of the trivial fact that the global benchmark $\mathbf{p}^*$ is an element of the set over which the epoch-local optimum is taken. Chaining Equation (11) with Equation (12),

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i. \end{aligned} \quad (13)$$

Step 5 (Per-epoch combined bound). Chaining Equations (10) and (13), for every epoch $i \in \{1, \dots, K'\}$ we obtain

$$\begin{aligned} & \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (14)$$

Equation (14) is the per-epoch regret bound *against the global benchmark* $\mathbf{p}^*$, and it is the building block for the rest of the proof. Note that despite the fact that $\mathbf{p}^*$ is the benchmark over all T rounds (not just epoch i), we are able to bound our per-epoch regret against $\mathbf{p}^*$ because $\mathbf{p}^* \in \mathcal{P}$ and we applied Lemma 1 on the sub-sequence.

Step 6 (Discovery rounds). For each discovery round $t \in \mathcal{D}$, the algorithm either plays $\mathbf{p}_{\text{def}}$ (if $t = \tau_1$) or samples $\mathbf{p}_t$ from the PW distribution inherited from Ep_{i-1} (if $t = \tau_i$ for $i \geq 2$). In both cases the sampled $\mathbf{p}_t$ is not tuned to the newly-appearing type $a_t = a_{\tau_i}$. However, because the payoff U_d is bounded in $[-1, 1]$, for every $\mathbf{p} \in \mathcal{P}$ and every type α , $U_d(b_\alpha(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so

$$|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)| \leq 2. \quad (15)$$

Summing Equation (15) over the at most $K' \leq K_{\max}$ discovery rounds yields

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq 2K_{\max}. \quad (16)$$

This is the only source of additive dependence on $K_{\max}$ that does not come from the main $\sqrt{TK_{\max}}$ term. It is the “price of not knowing K ” at a discovery round.

Step 7 (Uniform bound on $|\mathcal{E}_i|$). By Lemma 2 applied with $|C_i| \leq K_{\max}$,

$$|\mathcal{E}_i| \leq c_2 (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} =: N_{\max},$$

for an absolute constant c_2 . Taking logarithms,

$$\begin{aligned} \log N_{\max} &= n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\leq n^2 + n \log(1 + \frac{K_{\max} n^2}{2^{n-1}}) + K_{\max} \log n \\ &= O(n^2 + K_{\max} \log n), \end{aligned} \quad (17)$$

where the middle step uses $\log(2^n + K_{\max} n^2) \leq n + \log(1 + K_{\max} n^2 / 2^{n-1})$ and $\log(1 + x) \leq x$ for $x > 0$. Crucially, Equation (17) is uniform in i : $\log |\mathcal{E}_i| \leq \log N_{\max}$ for every epoch. This lets us pull the $\log |\mathcal{E}_i|$ factor in Step 5 outside the sum over epochs.

Step 8 (Summing across epochs). Sum Equation (14) over $i = 1, \dots, K'$ and add Equation (16). The algorithm’s total expected utility is then at least the benchmark’s total utility, minus the discovery term $2K_{\max}$, minus $2\epsilon T$ from the telescoping Lemma-4.3 approximation errors (using $\sum_i L_i \leq T$), minus the sum of the per-epoch PW regret terms. Rearranging,

$$\mathbb{E}[\text{Regret}(T)] \leq 2K_{\max} + 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=1}^{K'} \sqrt{L_i}. \quad (18)$$

Step 9 (Cauchy–Schwarz). The last term involves the sum of K' square roots. By the Cauchy–Schwarz inequality applied to $(1, \dots, 1) \in \mathbb{R}^{K'}$ and $(\sqrt{L_1}, \dots, \sqrt{L_{K'}})$,

$$\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \leq \sqrt{K_{\max} T}. \quad (19)$$

The first inequality is tight when all epochs have equal length, a scenario that corresponds to an adversary who reveals types evenly across the horizon. This is where the leading multiplicative $\sqrt{K_{\max}}$ factor in the bound is generated. An instance-adaptive version uses K' in place of $K_{\max}$, which is what Corollary 8 records.

Step 10 (Final substitution). Plugging Equations (17) and (19) into Equation (18) and choosing $\epsilon = 1/\sqrt{T}$ gives

$$\begin{aligned} \mathbb{E}[\text{Regret}(T)] &\leq 2K_{\max} + 2\sqrt{T} + c_1 \sqrt{K_{\max} T \log N_{\max}} \\ &\leq 2K_{\max} + 2\sqrt{T} \\ &\quad + c_3 \sqrt{TK_{\max}(n^2 + K_{\max} \log n)} \end{aligned}$$

for an absolute constant c_3 . Whenever $T \geq K_{\max}$ (the only regime in which the statement of Theorem 6 is non-trivial) the leading $\sqrt{TK_{\max}(n^2 + K_{\max} \log n)}$ term dominates the additive $2K_{\max} + 2\sqrt{T}$, so

$$\mathbb{E}[\text{Regret}(T)] = O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right),$$

which is exactly the bound of Theorem 6. This completes the proof. $\square$

6.1 Consequences of the proof

Two corollaries follow directly from the proof of Theorem 6 above. The first characterizes the regime in which the bound is non-trivial, the second gives an instance-adaptive refinement.

Corollary 7 (Regime of applicability). *The bound of Theorem 6 is sublinear in T if and only if $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$; equivalently, $K_{\max} = o(\min\{T/n^2, \sqrt{T/\log n}\})$. In particular, for $K_{\max} = O(1)$ the rate is $\tilde{O}(n\sqrt{T})$.*

Proof. Set the right-hand side of Equation (7) equal to T and solve for $K_{\max}$. The condition $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$ is exactly when the bound of Theorem 6 is $o(T)$; the equivalent form follows by separately bounding $K_{\max} \cdot n^2 \leq T/\omega(1)$ and $K_{\max}^2 \log n \leq T/\omega(1)$. $\square$

Corollary 8 (Instance-adaptive bound). *Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ be the number of distinct attacker types actually observed in the realized sequence. Then Algorithm 1 achieves*

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K' \cdot (n^2 + K' \log n)}\right).$$

Proof. The Cauchy–Schwarz step Equation (19) of the proof of Theorem 6 gives $\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot T}$; plugging this sharper bound into Equation (18) and redoing Step 10 with K' in place of $K_{\max}$ gives the stated bound. $\square$

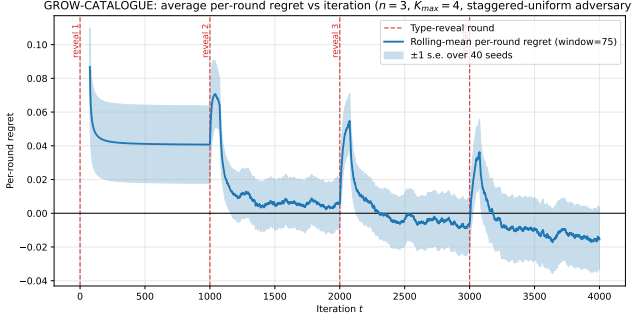


Figure 1: Rolling-window per-round regret of GROW-CATALOGUE under the staggered-uniform adversary ($n = 3$, $K_{\max} = 4$, $T = 4000$, average over 40 seeds, window = 75). Red dashed lines mark the discovery rounds 0, 1000, 2000, 3000. Each reveal causes a visible transient spike, followed by PW re-convergence inside the new epoch.

7 Experiments

7.1 Setup

We implement Algorithm 1 in Python, along with two reference algorithms: the known- K oracle of Balcan et al. [2, Sec. 5], which receives the full set K at $t = 1$; and a *uniform* baseline that plays $\mathbf{p}_{\text{def}} = (R/n, \dots, R/n)$ every round. Unless stated otherwise, we use $n = 3$ targets, $R = 1$ resource, and attacker utilities drawn i.i.d. from $\text{Unif}[-1, 1]$ with $u_{\alpha}^u(i) > 0 > u_{\alpha}^c(i)$ to ensure covering is costly for the attacker. The extreme-point set $\mathcal{E}(C; \epsilon)$ is computed by direct enumeration of intersections of $(n - 1)$ -subsets of the defining hyperplanes, which is tractable at $n = 3$.

We use a *staggered-uniform* adversary: the horizon $[0, T]$ is partitioned into $K_{\max}$ phases of length $T/K_{\max}$ each, and in phase i the attacker is drawn uniformly at random from the first i types $\{\alpha_1, \dots, \alpha_i\}$. Thus α_1 is present in every phase, α_2 from phase 2 onward, and so on. This design produces a clean sequence of discovery rounds at $t = 0, T/K_{\max}, \dots, (K_{\max} - 1)T/K_{\max}$, while still mixing old and new types after each reveal so that the benchmark $\mathbf{p}^*$ is a genuine compromise rather than a specialist strategy.

7.2 Results

Figure 1 shows that the algorithm’s per-round regret tracks the epoch structure of the proof: each discovery round produces a bounded transient, and the PW sub-routine re-converges within the new epoch. The spikes shrink with i because, by epoch i , the algorithm already has a good PW distribution on most of the new (refined) extreme points.

Figure 2 plots cumulative $\text{Regret}(T)$ versus T with the $C\sqrt{T}$ envelope from Theorem 6 overlaid. The empirical regret stays visibly below the envelope for all T tested, and the ratio $\text{Regret}(T)/\sqrt{T}$ remains bounded (in fact, it decreases, indicating that the stochastic staggered-uniform adversary is far from the worst case). This is entirely consistent with Theorem 6, which is our upper bound.

Table 1 compares the four algorithms at $T = 4000$. The regret of GROW-CATALOGUE is within roughly 15%–40% of

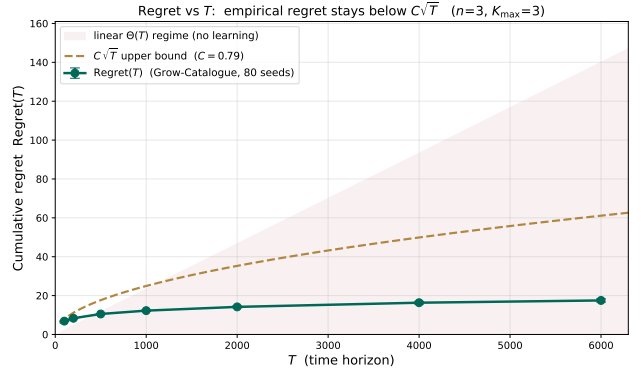


Figure 2: Cumulative regret versus horizon T for GROW-CATALOGUE ($n = 3$, $K_{\max} = 3$, uniform random adversary, 80 seeds). The green curve stays strictly below the $C\sqrt{T}$ upper envelope implied by Theorem 6, which is in turn well below the linear-regret regime that would result without learning.

the known- K oracle and an order of magnitude better than the uniform baseline. The two GROW-CATALOGUE variants produce regret that differs only by Monte-Carlo noise, as predicted by the observational equivalence of the two recomputation schemes.

Algorithm	$\mathbb{E}[\text{Regret}(T)]$	Ratio
Known- K oracle of Balcan et al. [2]	14.51 ± 5.00	1.00
GROW-CAT. (total)	16.57 ± 5.40	1.14
Uniform baseline	292.77 ± 350.28	20.18

Table 1: Cumulative regret at $T = 4000$, $n = 3$, $K_{\max} = 3$, uniform-random adversary, mean $\pm$ std over 15 seeds. The known- K oracle receives K in advance; GROW-CATALOGUE attains comparable regret *without* this information.

8 Conclusions

We studied the repeated Stackelberg security game in a new, open regime in which the set of attacker types is unknown to the defender a priori but known to have bounded cardinality. We presented GROW-CATALOGUE, a simple online algorithm that incrementally discovers attacker types and re-partitions the coverage polytope accordingly, and we gave a full proof that its expected regret under full-information feedback is at most $O(\sqrt{T} \cdot K_{\max} \cdot (n^2 + K_{\max} \log n))$. This is the same functional form as the known- K bound of Balcan et al. [2, Theorem 5.1], with the known cardinality k replaced by our known upper bound $K_{\max}$. The bound is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$, and the matching $\Omega(T)$ lower bound of Balcan et al. [2, Theorem 7.1] rules out further improvements on the unbounded side. Our experiments confirm the analytical scaling: the empirical regret stays below the $C\sqrt{T}$ envelope and tracks the epoch structure of the proof.

9 Extensions

This section reports work carried out after the version of the paper that ends with Section 8. It has three parts. Section 9.1 corrects an unjustified approximation in Step 7 of the proof of Theorem 6; the statement of Theorem 6 is unaffected, but the displayed derivation that supports it was not valid as written, and the correction also forces us to restate more carefully how our rate compares with that of Balcan et al. [2]. Section 9.2 generalises the information model: instead of learning a new type at the instant it first attacks, the defender receives the type catalogue in periodic instalments, every P rounds. Section 9.3 generalises in an orthogonal direction, by adding side information in the sense of Harris et al. [9], and asks what survives when K is unknown.

Throughout this section we write

$$\Lambda := n^2 + K_{\max} \log n, \quad (20)$$

so that Theorem 6 reads $\mathbb{E}[\text{Regret}(T)] = O(\sqrt{T K_{\max} \Lambda})$. All logarithms are base 2.

9.1 Erratum: Step 7 of the proof of Theorem 6

Step 7 of Section 6 bounds $\log N_{\max}$ by the chain

$$\begin{aligned} \log N_{\max} &= n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\leq n^2 + n \log\left(1 + \frac{K_{\max} n^2}{2^n - 1}\right) + K_{\max} \log n, \end{aligned}$$

and then asserts that the right-hand side is $O(n^2 + K_{\max} \log n)$, “using $\log(1 + x) \leq x$ for $x > 0$ ”. That last step is not valid as written, for two separate reasons.

1. The inequality $\log(1 + x) \leq x$ is a statement about the *natural* logarithm. The same display uses $\log 2^n = n$, i.e. base 2, and $\log_2(1 + x) > x$ for every $x \in (0, 1)$.
2. Even granting the inequality, it is applied exactly where it is least informative. It yields the bound $K_{\max} n^3 / 2^{n-1}$ for a quantity whose true size is logarithmic in $K_{\max}$. For $n = 2$ and $K_{\max} = 10^6$, for instance, $n \log(2^n + K_{\max} n^2) = 43.9$ whereas the substitute expression $n^2 + K_{\max} n^3 / 2^{n-1}$ equals 4.0×10^6 — an overestimate by a factor of about 9×10^4 . A first-order expansion of the logarithm is simply the wrong tool for a term in which the argument of the logarithm is unbounded.

Lemma 9 restores the conclusion by the route actually taken by Balcan et al. [2, proof of Theorem 5.1], namely bounding a sum inside a logarithm by its larger summand rather than expanding it.

Lemma 9 (Uniform expert-set bound; corrected Step 7). *Let $n \geq 2$ and $K_{\max} \geq 2$. For every catalogue $C \subseteq K$ with $|C| \leq K_{\max}$ and every $\epsilon > 0$,*

$$\begin{aligned} \log |\mathcal{E}(C; \epsilon)| &\leq \log N_{\max} \leq 5n^2 + 2K_{\max} \log n + O(1) \\ &= O(n^2 + K_{\max} \log n), \end{aligned}$$

where $N_{\max} = c_2(2^n + K_{\max} n^2)^n n^{K_{\max}}$ is as in Step 7.

Proof. The first inequality is Lemma 2 together with the monotonicity of the bound in $|C|$. Taking logarithms,

$$\log N_{\max} = n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1). \quad (21)$$

For any $a, b > 0$ we have $a + b \leq 2 \max\{a, b\}$ and hence $\log(a + b) \leq 1 + \max\{\log a, \log b\}$. Applying this with $a = 2^n$ and $b = K_{\max} n^2$,

$$\begin{aligned} \log(2^n + K_{\max} n^2) &\leq 1 + \max\{n, \log K_{\max} + 2 \log n\} \\ &\leq 1 + n + \log K_{\max} + 2 \log n, \end{aligned} \quad (22)$$

so that $n \log(2^n + K_{\max} n^2) \leq n + n^2 + n \log K_{\max} + 2n \log n$. We claim that

$$n \log K_{\max} \leq n^2 + K_{\max} \log n. \quad (23)$$

Indeed, if $K_{\max} \leq 2^n$ then $n \log K_{\max} \leq n \cdot n = n^2$. If $K_{\max} > 2^n$, consider $f(x) = x / \log x$, which is increasing on $[e, \infty)$. For $n \geq 3$ we have $e \leq n < 2^n < K_{\max}$, so $f(K_{\max}) \geq f(n)$, i.e. $n \log K_{\max} \leq K_{\max} \log n$; for $n = 2$ the hypothesis $K_{\max} > 2^n = 4$ gives $K_{\max} \geq 5$ and $f(K_{\max}) \geq f(5) = 5 / \log 5 > 2 = f(2)$, so the same conclusion holds. This proves Equation (23). Using $2n \log n \leq 2n^2$ and $n \leq n^2$ and substituting into Equation (21),

$$\log N_{\max} \leq 5n^2 + 2K_{\max} \log n + O(1),$$

which is the claim: the five n^2 terms come from n , n^2 , the n^2 of Equation (23) and $2n \log n \leq 2n^2$, and the two $K_{\max} \log n$ terms from Equation (23) and Equation (21). $\square$

Lemma 9 delivers exactly Equation (17), so Steps 8–10 of Section 6, Theorem 6, Corollary 7 and Corollary 8 all stand unchanged. Only the justification inside Step 7 needs replacing.

Remark 10 (On the comparison with Balcan et al.). The corrected computation also makes visible a second inaccuracy, in the *prose* around Theorem 6: the abstract, Theorem 6 and Section 8 describe our rate as matching Balcan et al. [2, Theorem 5.1] “with k replaced by $K_{\max}$ ”. That is not right. The bound displayed in Balcan et al. [2, Theorem 5.1], $O(\sqrt{T n^2 k \log n k})$, is itself a loosening; what their proof establishes is $O(\sqrt{T \log |\mathcal{E}|})$, and by exactly the computation of Lemma 9 with $K_{\max}$ replaced by k this equals

$$O\left(\sqrt{T(n^2 + k \log n)}\right) \quad (\text{sharp known-}K \text{ rate}).$$

Written against this sharp form, Theorem 6 is $O(\sqrt{T K_{\max} \Lambda})$ versus $O(\sqrt{T \Lambda})$ with $k = K_{\max}$: a multiplicative $\sqrt{K_{\max}}$ worse, generated entirely by the Cauchy–Schwarz step (Step 9) over the $K' \leq K_{\max}$ epochs. That factor is the price of not knowing K , and it should be stated as such rather than described as a match. The claim “for $K_{\max} = O(1)$ the rate is $\tilde{O}(n\sqrt{T})$ ” is correct, since then $K_{\max}$ is absorbed into the constant.

9.2 Direction 1: periodic type revelation

Problem 4 couples two things that need not be coupled: the moment a type *acts* and the moment the defender *learns what that type wants*. In deployment the second is usually the output of an intelligence process that reports on a schedule — a weekly threat briefing, a monthly poaching-patrol debrief — rather than instantaneously. We therefore study the model in which the catalogue is delivered in instalments of period P .

9.2.1 Model and algorithm

Problem 11 (Periodic revelation, PR-P). Fix a period $P \in \{1, \dots, T\}$ and partition the horizon into $B = \lceil T/P \rceil$ blocks $\mathcal{B}_j = \{(j-1)P + 1, \dots, \min(jP, T)\}$. A repeated SSG as in Section 3 is played with $|K| \leq K_{\max}$ and K unknown, and with the following feedback.

1. After each round t the defender observes an *identifier* for a_t : enough to decide whether a_t has been encountered before, but nothing about its payoffs.
2. At each *checkpoint* $t = jP$, $j = 1, \dots, B$, the defender is given the utility vectors (u_α^c, u_α^u) of every type that has appeared in rounds $1, \dots, jP$.

Thus the defender's *catalogue* at round t is $C_t = \{a_s : s \leq P \lfloor (t-1)/P \rfloor\}$, which is constant on each block. The benchmark $\mathbf{p}^*$ and $\text{Regret}(T)$ are as in Equations (3) and (4).

Taking $P = 1$ recovers Problem 4 exactly: the checkpoint at the end of round t discloses a_t 's dossier, so $C_t = \{a_1, \dots, a_{t-1}\}$, which is the catalogue Algorithm 1 holds when it plays $\mathbf{p}_t$. Increasing P batches the same information into fewer, larger deliveries.

Two features of PR-P drive the analysis.

Deterministic epoch boundaries. The catalogue can change only at a checkpoint, so the epoch structure of Section 6 is now anchored to a grid known to the defender in advance. Write $D_j = \{a_t : t \leq jP\}$ for the types instantiated by the end of block j (so $D_0 = \emptyset$ and block j is played with catalogue D_{j-1}), and let

$$G = \{j \in [B] : D_j \supsetneq D_{j-1}\}, \quad M := |G|.$$

Every $j \in G$ contributes at least one type never seen before, so

$$M \leq \min\{K', B\} \leq \min\{K_{\max}, \lceil T/P \rceil\}. \quad (24)$$

We write $\overline{M} := \min\{K_{\max}, \lceil T/P \rceil\}$ for the right-hand side of Equation (24), a quantity the defender knows in advance. The catalogue changes only at the starts of the blocks $j+1$ with $j \in G$, so the horizon splits into at most $M+1$ *epochs* on which the catalogue — and hence the expert set — is constant.

Blind rounds. Call round t *blind* if $a_t \notin C_t$. On a blind round the defender cannot evaluate $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$ for $\mathbf{p} \in \mathcal{E}$, because it does not hold a_t 's payoffs, and so cannot update its weights. This is the one genuinely new cost in PR-P; Lemma 12 bounds how many such rounds there can be.

Algorithm 2 is Algorithm 1 with the catalogue update moved onto the checkpoint grid, and with one deliberate change: the Polynomial-Weights sub-routine is restarted *lazily*, only when the catalogue actually grows, and not at every block boundary. Remark 14 below shows how much that matters.

Algorithm 2 PERIODIC-GROW-CATALOGUE (PR-P).

Require: horizon T , period P , game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

```

1:  $C \leftarrow \emptyset$ ;  $\mathcal{E} \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/\sqrt{T}$ ;  $t_{\text{loc}} \leftarrow 0$ 
2: for  $t = 1, \dots, T$  do
3:   if  $\mathcal{E} = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     sample  $\mathbf{p}_t \sim q_t$ , the current PW distribution over  $\mathcal{E}$ 
7:   end if
8:   observe the identifier of  $a_t$ 
9:   if  $a_t \in C$  then ▷ informed round
10:    compute  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$  for  $\mathbf{p} \in \mathcal{E}$ 
11:     $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$ ;  $\eta \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$ 
12:     $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} e^{-\eta \ell_t(\mathbf{p})}$  for all  $\mathbf{p} \in \mathcal{E}$ 
13:   else ▷ blind round: freeze the weights
14:   end if
15:   if  $t \equiv 0 \pmod{P}$  then ▷ checkpoint
16:    receive the dossiers of all types seen so far;  $C' \leftarrow \{a_s : s \leq t\}$ 
17:    if  $C' \neq C$  then ▷ lazy restart
18:       $C \leftarrow C'$ ;  $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$ ; reset PW on  $\mathcal{E}$ ;  $t_{\text{loc}} \leftarrow 0$ 
19:    end if
20:   end if
21: end for
```

9.2.2 Regret analysis

Lemma 12 (Blind rounds). *Let $\bar{S} = \{t \in [T] : a_t \notin C_t\}$ be the set of blind rounds. Then $|\bar{S}| \leq \min\{MP, T\}$, and*

$$\sum_{t \in \bar{S}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in \bar{S}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq 2 \min\{MP, T\}.$$

Proof. Let $t \in \bar{S}$. Then $C_t = D_{j-1}$, so t is blind exactly when a_t first appears during block j , which forces $D_j \supsetneq D_{j-1}$, i.e. $j \in G$. Hence $\bar{S} \subseteq \bigcup_{j \in G} \mathcal{B}_j$ and $|\bar{S}| \leq |G| \cdot P = MP$; the bound $|\bar{S}| \leq T$ is trivial. The second claim follows because $U_d(i, \mathbf{p}) \in [-1, 1]$ for every target and every $\mathbf{p} \in \mathcal{P}$, so each round contributes at most 2. $\square$

Theorem 13 (Regret under periodic revelation). *Let the defender play Algorithm 2 in the setting of Problem 11. For any oblivious adversarial sequence $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$,*

$$\mathbb{E}[\text{Regret}(T)] \leq 2\sqrt{T} + c_1 \sqrt{(M+1)T \log N_{\max}} + 2 \min\{MP, T\}, \quad (25)$$

with $M \leq \overline{M} = \min\{K_{\max}, \lceil T/P \rceil\}$ as in Equation (24) and c_1 the constant of Proposition 3. Consequently, with Λ as in Equation (20),

$$\mathbb{E}[\text{Regret}(T)] = O\left(\sqrt{T \overline{M} \Lambda} + \overline{M} P\right). \quad (26)$$

Proof. Fix the sequence $\mathbf{a}$ and let $\text{Ep}_0, \dots, \text{Ep}_{M'}, M' \leq M$, be the epochs on which the catalogue is constant, with catalogues $C^{(i)}$, expert sets $\mathcal{E}_i = \mathcal{E}(C^{(i)}; \epsilon)$ and lengths L_i , $\sum_i L_i = T$. Split each epoch into its informed and blind rounds, $\text{Ep}_i = S_i \sqcup \bar{S}_i$, and put $\ell_i = |S_i|$.

Step 1 (within-epoch regret on the informed subsequence). Algorithm 2 feeds the Polynomial-Weights sub-routine exactly

the loss vectors of the rounds in S_i , in order, over the fixed expert set $\mathcal{E}_i$, with losses in $[-1, 1]$. Proposition 3 with $\kappa = 1$ and horizon ℓ_i therefore gives

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in S_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[\sum_{t \in S_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq c_1 \sqrt{\ell_i \log |\mathcal{E}_i|}. \quad (27)$$

Freezing the weights on $\bar{S}_i$ is invisible to this bound: the sub-routine simply never sees those rounds.

Step 2 (passage to $\mathbf{p}^$).* By construction every $t \in S_i$ has $a_t \in C^{(i)}$, which is precisely the hypothesis of Lemma 1. Applying it to the catalogue $C^{(i)}$ and the sub-sequence $(a_t)_{t \in S_i}$ of length ℓ_i , and then using $\mathbf{p}^* \in \mathcal{P}$ as a feasible competitor exactly as in Step 4b of Section 6,

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in S_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \geq \sum_{t \in S_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon \ell_i. \quad (28)$$

Step 3 (assembling). Chaining Equations (27) and (28), summing over $i = 0, \dots, M'$, adding Lemma 12 for the blind rounds and using $\sum_i \ell_i \leq T$,

$$\mathbb{E}[\text{Regret}(T)] \leq 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=0}^{M'} \sqrt{\ell_i} + 2 \min\{MP, T\},$$

where $\log |\mathcal{E}_i| \leq \log N_{\max}$ uniformly by Lemma 9. By Cauchy–Schwarz, $\sum_{i=0}^{M'} \sqrt{\ell_i} \leq \sqrt{(M'+1)T} \leq \sqrt{(M+1)T}$, and choosing $\epsilon = 1/\sqrt{T}$ gives Equation (25). Equation (26) follows from Equation (24) and Lemma 9. $\square$

Setting $P = 1$ in Equation (26) gives $M \leq K_{\max}$ and $MP \leq K_{\max}$, i.e. $O(\sqrt{TK_{\max}\Lambda} + K_{\max})$, which is Theorem 6: the analysis degenerates to the original one, as it must.

Remark 14 (Why the restart must be lazy). The first analysis one is tempted to write treats every block as an epoch, because the block boundaries are exactly the points at which the defender *could* learn something. That gives $B = \lceil T/P \rceil$ epochs of length P and, through Step 9, $\sum_{j=1}^B \sqrt{P} = T/\sqrt{P}$, hence a bound of the shape

$$O\left(\sqrt{T \cdot (T/P) \cdot \Lambda}\right).$$

This is correct but wasteful. Restarting Polynomial Weights on a block whose catalogue is unchanged throws away a converged weight vector for nothing, and the count T/P ignores the fact that the catalogue can grow at most $K_{\max}$ times in total. The lazy restart of Algorithm 2 replaces T/P by $\min\{K_{\max}, T/P\}$, which is never worse and is better by a factor $\sqrt{T/(PK_{\max})}$ whenever $P < T/K_{\max}$. The genuinely new term in Equation (26) is not the epoch count at all — it is the blind-round term MP .

Corollary 15 (Optimal period). *Assume the regime of Corollary 7, i.e. $K_{\max}\Lambda \leq T$, and set*

$$P^* := \sqrt{\frac{T\Lambda}{K_{\max}}} = \sqrt{\frac{T(n^2 + K_{\max} \log n)}{K_{\max}}}.$$

Then $P^ \leq T/K_{\max}$, and:*

1. (Batching is free.) *For every $P \leq P^*$,*

$$\mathbb{E}[\text{Regret}(T)] = O\left(\sqrt{T K_{\max} \Lambda}\right),$$

the rate of Theorem 6. Periodic revelation with any period up to P^ costs nothing asymptotically.*

2. (No-regret threshold.) *Algorithm 2 is a no-regret algorithm whenever $P = o(T/K_{\max})$, and Equation (26) becomes vacuous (linear in T) once $P = \Omega(T/K_{\max})$.*

Proof. $P^* \leq T/K_{\max}$ is equivalent to $T\Lambda/K_{\max} \leq T^2/K_{\max}^2$, i.e. to $K_{\max}\Lambda \leq T$, which is the standing assumption. For (i), $P \leq P^* \leq T/K_{\max}$ gives $M \leq K_{\max}$ and $2MP \leq 2K_{\max}P^* = 2\sqrt{TK_{\max}\Lambda}$, so the blind-round term in Equation (26) is dominated by the learning term. For (ii), when $P \leq T/K_{\max}$ we have $M \leq K_{\max}$ and the bound is $O(\sqrt{TK_{\max}\Lambda} + K_{\max}P)$, which is $o(T)$ iff $K_{\max}P = o(T)$, using $K_{\max}\Lambda = o(T)$ for the first term; when $P \geq T/K_{\max}$ we have $M = \lceil T/P \rceil$ and $MP \geq T$, so Equation (26) exceeds T . $\square$

Two readings of Corollary 15 are worth separating. If P is a fact about the environment, the corollary says how much staleness the defender can tolerate before the analysis stops being informative: intelligence delivered at least once every $\Theta(n\sqrt{T/K_{\max}})$ rounds (the value of P^* when $K_{\max} \lesssim n^2/\log n$) is as good as instantaneous. If instead P is a design parameter — how often to convene the analysts — the corollary says there is no reason to convene more often than every P^* rounds, because the regret bound is flat below that.

9.2.3 A lower bound: batching strengthens the adversary

Corollary 15(ii) is an upper-bound statement, and one should ask whether the degeneration at $P = \Omega(T/K_{\max})$ is an artefact. It is not. Theorem 16 shows that batching transplants the hardness of Balcan et al. [2, Theorem 7.1] from round granularity to block granularity, and in doing so it slashes the number of attacker types the adversary needs in order to force linear regret.

Theorem 16 (Lower bound under periodic revelation). *Fix $P \geq 1$ and $T \geq 4P$, and let $B = \lfloor T/P \rfloor$. There is a PR- P repeated security game with $n = 3$ targets, one resource, and a set K of at most $2(8P)^B$ attacker types on which every online algorithm — deterministic or randomised — has expected regret at least $T/2$.*

Proof. We use the game of Balcan et al. [2, Theorem 7.1]: $N = \{1, 2, 3\}$, one resource, $U_d(1, \mathbf{p}) = U_d(3, \mathbf{p}) = -1$ and $U_d(2, \mathbf{p}) = 0$. Because the defender strictly prefers target 2 to be attacked, setting $p_2 = 0$ never hurts, so we may restrict attention to $\mathbf{p} = [p, 0, 1-p]$ with $p \in [0, 1]$ and identify a strategy with the scalar p . By Balcan et al. [2, Lemma 7.2], for any $0 \leq r_1 \leq r_2 \leq 1$ there is an attacker type α whose best response is target 1 on $[0, r_1]$, target 2 on $(r_1, r_2]$ and target 3 on $(r_2, 1]$; the defender's payoff against α is therefore 0 if $p \in (r_1, r_2]$ and -1 otherwise.

Let $c = \lceil \log_2(4P) \rceil$, so that $4P \leq 2^c \leq 8P$. Build a 2^c -ary tree of depth B whose level- j nodes are the 2^{j^c} dyadic sub-intervals of $(0, 1]$ of length 2^{-j^c} , each node's children

partitioning it. Associate with a node v , carrying interval $I_v = (l_v, r_v]$, the type α_v obtained from Balcan et al.'s Lemma 7.2 with $r_1 = l_v$, $r_2 = r_v$. The total number of types is $\sum_{j=1}^B 2^{jc} \leq 2 \cdot 2^{Bc} \leq 2(8P)^B$.

The adversary. Draw a uniformly random root-to-leaf path $v_1, \dots, v_B$, each v_j uniform among the 2^c children of v_{j-1} , and play α_{v_j} at every round of block $\mathcal{B}_j$ (and α_{v_B} in any leftover rounds). The identifiers of the level- j types are permuted uniformly at random, so an identifier carries no information about which child was chosen.

The benchmark. The intervals are nested, $I_{v_B} \subseteq \dots \subseteq I_{v_1}$. Any $\mathbf{p}^*$ with $\mathbf{p}^* \in I_{v_B}$ therefore satisfies $b_{\alpha_{v_j}}(\mathbf{p}^*) = 2$ for every j , so $\mathbf{p}^*$ collects utility 0 in every one of the T rounds.

The defender's information inside block j . At the start of block j the checkpoints have disclosed the dossiers of $\alpha_{v_1}, \dots, \alpha_{v_{j-1}}$, hence $I_{v_{j-1}}$, but nothing about α_{v_j} . During the block the only information the defender receives is its own payoff, which equals 0 if $p_t \in I_{v_j}$ and -1 otherwise; it does not distinguish target 1 from target 3. So each round reveals a single bit: whether the strategy played landed in the unknown child interval.

Failure probability. Condition on the history H_r after r rounds of block j in which every bit was 0. Each round's p_t lies in at most one child of $I_{v_{j-1}}$, so at most r children have been excluded, and the posterior law of v_j given H_r is uniform on the remaining $2^c - r$ children — and independent of the algorithm's internal randomness at round $r + 1$. Hence $\Pr[\text{round } r + 1 \text{ succeeds} \mid H_r] \leq 1/(2^c - r)$, and

$$\begin{aligned} \Pr[\exists t \in \mathcal{B}_j : p_t \in I_{v_j}] &\leq \sum_{r=0}^{P-1} \frac{1}{2^c - r} \\ &\leq \frac{P}{2^c - P} \leq \frac{P}{3P} = \frac{1}{3}. \end{aligned}$$

On the complementary event every one of the $|\mathcal{B}_j|$ rounds of the block pays -1 , so the expected penalty in block j is at least $\frac{2}{3}|\mathcal{B}_j|$.

Conclusion. Summing over the B blocks, the expected total penalty is at least $\frac{2}{3}BP \geq \frac{2}{3}(T - P) \geq \frac{2}{3} \cdot \frac{3}{4}T = \frac{T}{2}$, using $T \geq 4P$. Since $\mathbf{p}^*$ has utility 0, the expected regret is at least $T/2$, the expectation being over the adversary's randomisation and the algorithm's. By Yao's minimax principle there is a deterministic choice of path achieving at least this value against the given randomised algorithm. $\square$

Remark 17 (How much batching costs the defender). Theorem 16 is best read as a statement about the adversary's budget. To force regret $T/2$ against an immediately-informed defender, Balcan et al. [2, Theorem 7.1] need 2^{T+1} attacker types. Under PR- P the same regret is forced with $2(8P)^{T/P} = 2^{\Theta((T/P) \log P)}$ types — for $P = \sqrt{T}$, that is $2^{\Theta(\sqrt{T} \log T)}$ instead of $2^{\Theta(T)}$. Batching does not merely delay information; it makes a far smaller type set sufficient to defeat any learner. Setting $P = 1$ recovers a (slightly lossier) form of Balcan et al.'s bound, as it should.

We do *not* claim that the blind-round term MP of Equation (26) is tight for small $K_{\max}$: Theorem 16 needs a type budget exponential in B , whereas the upper bound is stated for $|K| \leq K_{\max}$. Closing that gap — in particular, deciding whether $\Theta(K_{\max}P)$ is the right blind-round cost when $K_{\max}$ is a constant — is left open; see Section 9.4.

9.3 Direction 2: side information with an unknown type set

Harris et al. [9] add *side information* to the Stackelberg model: before the leader commits, both players observe a context $z_t \in \mathcal{Z}$, and both players' payoffs may depend on it. Their benchmark is the best *policy* in hindsight, a map from contexts to strategies. They show that no-regret learning is impossible when contexts and follower types are both adversarial, and recover no-regret under either of two relaxations. Their entire development assumes, as Balcan et al. [2] do, that the set of K follower types is known in advance. This subsection asks what happens when it is not.

9.3.1 Model

Problem 18 (Contextual unknown- K repeated SSG). A context space $\mathcal{Z} \subseteq \mathbb{R}^d$ is fixed. At each round $t \in \{1, \dots, T\}$:

1. Nature reveals $z_t \in \mathcal{Z}$ to both players;
2. the defender commits to $\mathbf{p}_t \in \mathcal{P}$, possibly drawn from a *policy* $\pi_t : \mathcal{Z} \rightarrow \mathcal{P}$;
3. attacker a_t observes $(z_t, \mathbf{p}_t)$ and attacks $b_{a_t}(z_t, \mathbf{p}_t) = \arg \max_{i \in N} U_{a_t}(i, \mathbf{p}_t; z_t)$;
4. the defender receives $U_d(b_{a_t}(z_t, \mathbf{p}_t), \mathbf{p}_t; z_t)$ and observes a_t ; if a_t has not been seen before, its utility functions are revealed.

Payoffs keep the form of Equation (1) at each context: $U_d(i, \mathbf{p}; z) = u_d^c(z, i) p_i + u_d^u(z, i)(1 - p_i)$ and $U_{a_t}(i, \mathbf{p}; z) = u_{a_t}^c(z, i) p_i + u_{a_t}^u(z, i)(1 - p_i)$, so both are linear in $\mathbf{p}$ for each fixed (i, z) . Following Harris et al. [9] we normalise $U_d \in [0, 1]$ rather than $[-1, 1]$; this only rescales constants. The type set K is unknown with $|K| \leq K_{\max}$. The benchmark is the best policy in hindsight,

$$\pi^*(z) \in \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^T U_d(b_{a_t}(z, \mathbf{p}), \mathbf{p}; z) \mathbf{1}\{z_t = z\},$$

and the *contextual Stackelberg regret* is $R(T) = \sum_{t=1}^T U_d(b_{a_t}(z_t, \pi^*(z_t)), \pi^*(z_t); z_t) - U_d(b_{a_t}(z_t, \mathbf{p}_t), \mathbf{p}_t; z_t)$.

9.3.2 Catalogue-indexed contextual best-response regions

The structural prerequisite for everything below is a catalogue-relative version of Harris et al.'s Definitions 4.1–4.3, i.e. the contextual analogue of Equations (5) and (6).

Definition 19 (Contextual best-response regions over a catalogue). Fix a context $z \in \mathcal{Z}$ and a catalogue $C \subseteq K$. For a type $\alpha \in C$ and a target $i \in N$ put $\mathcal{P}_i^\alpha(z) = \{\mathbf{p} \in \mathcal{P} : b_\alpha(z, \mathbf{p}) = i\}$, the intersection of $\mathcal{P}$ with the $n - 1$ linear inequalities $U_\alpha(i, \mathbf{p}; z) \geq U_\alpha(j, \mathbf{p}; z)$, $j \neq i$. For a response profile $\sigma : C \rightarrow N$ put $\mathcal{P}_\sigma(z; C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(z)$, and let $\Sigma_z(C)$ be the set of profiles with non-empty region. Each $\mathcal{P}_\sigma(z; C)$ is a bounded convex polytope, not necessarily closed, and $|\Sigma_z(C)| \leq n^{|C|}$. Finally, $\mathcal{E}_z(C; \delta)$ denotes the set of δ -approximate extreme points of the family $\{\mathcal{P}_\sigma(z; C)\}_{\sigma \in \Sigma_z(C)}$, built exactly as in Lemma 1: it contains every extreme point of $\text{cl}(\mathcal{P}_\sigma(z; C))$ that lies in $\mathcal{P}_\sigma(z; C)$, and otherwise a point of $\mathcal{P}_\sigma(z; C)$ within ℓ_1 -distance δ of it.

Two properties of Definition 19 are used repeatedly.

Lemma 20 (Refinement and a uniform size bound). *Fix $z \in \mathcal{Z}$ and catalogues $C \subseteq C' \subseteq K$ with $|C'| \leq K_{\max}$. Then*

1. (Refinement.) *Every cell $\mathcal{P}_{\sigma'}(z; C')$ is contained in the cell $\mathcal{P}_{\sigma'|_C}(z; C)$; hence the C' -partition of $\mathcal{P}$ refines the C -partition, and every extreme point of a C -cell is a vertex of the hyperplane arrangement defining the C' -partition.*
2. (Uniform size.) *For every $\delta > 0$, $\log |\mathcal{E}_z(C; \delta)| \leq \log N_{\max} = O(n^2 + K_{\max} \log n)$, uniformly in z and in C .*

Proof. (i) If $\mathbf{p} \in \mathcal{P}_{\sigma'}(z; C')$ then $b_{\alpha}(z, \mathbf{p}) = \sigma'(\alpha)$ for every $\alpha \in C' \supseteq C$, so $\mathbf{p} \in \mathcal{P}_{\sigma'|_C}(z; C)$. The half-spaces defining the C -partition form a subset of those defining the C' -partition, so any point that is the intersection of n linearly independent members of the former is also such an intersection of members of the latter.

(ii) At a fixed z the counting argument of Lemma 2 applies verbatim: the regions are cut out by the $O(2^n + n)$ half-spaces defining $\mathcal{P}$ together with $\binom{n}{2}$ indifference hyperplanes per type in C , so there are $O(2^n + n + |C|n^2)$ defining half-spaces and at most $O((2^n + |C|n^2)^n)$ extreme points, each spawning at most $n^{|C|}$ δ -approximate points. With $|C| \leq K_{\max}$ this is at most $N_{\max}$, and Lemma 9 gives the stated logarithm. The bound does not involve z . $\square$

Part (ii) is what makes the epoch machinery of Section 6 usable in the contextual setting: although the defender must recompute $\mathcal{E}_{z_t}(C_t; \delta)$ at every round (the context space is not assumed finite, so these sets cannot be precomputed), their *cardinality* is controlled uniformly by the same $N_{\max}$.

9.3.3 The fully adversarial setting remains unlearnable, even for $K_{\max} = 2$

In the non-contextual model, Remark 5 is the whole reason Problem 4 is interesting: Balcan et al. [2, Theorem 7.1] rule out no-regret learning when $|K|$ is unrestricted, and Theorem 6 shows that bounding $|K|$ by a known $K_{\max}$ restores it. The first thing to check in the contextual model is whether the same repair works. It does not, and it fails as badly as possible.

Theorem 21 (Impossibility with two attacker types). *There is a contextual repeated security game in the sense of Problem 18, with $n = 2$ targets, one resource, context space $\mathcal{Z} = [0, 1]$ and exactly two attacker types, on which every online algorithm suffers expected contextual Stackelberg regret $\Omega(T)$ when the sequences $z_1, \dots, z_T$ and $a_1, \dots, a_T$ are chosen adversarially. The bound holds even if the defender is told K , both dossiers and $K_{\max} = 2$ before the game starts.*

Proof. *The instance.* Take $N = \{1, 2\}$, one resource with schedules $\{\{1\}, \{2\}\}$, so $\mathcal{P} = \{\mathbf{p} \in [0, 1]^2 : p_1 + p_2 \leq 1\}$. Let the defender's payoffs be context-independent with $u_d^c(i) = 1$ and $u_d^u(i) = 0$, so that

$$U_d(i, \mathbf{p}) = p_i \in [0, 1] :$$

the defender scores 1 if the attacked target happens to be covered and 0 otherwise. Since U_d is non-decreasing in p_i and, as we now arrange, the attackers' best responses do not

depend on $\mathbf{p}$, we may assume $p_1 + p_2 = 1$ without loss of generality and write $g = p_1$.

Let α_1 have $u_{\alpha_1}^c(z, 1) = u_{\alpha_1}^u(z, 1) = 1$ and $u_{\alpha_1}^c(z, 2) = u_{\alpha_1}^u(z, 2) = 0$ for every z ; then $U_{\alpha_1}(1, \mathbf{p}; z) = 1 > 0 = U_{\alpha_1}(2, \mathbf{p}; z)$, so $b_{\alpha_1}(z, \mathbf{p}) = 1$ for all $z, \mathbf{p}$. Symmetrically let α_2 always attack target 2. Thus $K = \{\alpha_1, \alpha_2\}$ and $K_{\max} = 2$. The defender's realised payoff is

$$U_d(b_{\alpha_t}(z_t, \mathbf{p}_t), \mathbf{p}_t) = g_t \mathbf{1}\{a_t = \alpha_1\} + (1 - g_t) \mathbf{1}\{a_t = \alpha_2\}. \quad (29)$$

The reduction. Recall the online linear thresholding problem [9, Lemma 3.1]: an adversary fixes an unknown cutoff $s \in [0, 1]$; each round it reveals $\omega_t \in [0, 1]$, the learner outputs $g_t \in [0, 1]$, the label is $y_t = +1$ if $\omega_t > s$ and -1 otherwise, and the learner's utility is $u^{\text{OLT}}(\omega_t, g_t) = g_t \mathbf{1}\{y_t = 1\} + (1 - g_t) \mathbf{1}\{y_t = -1\}$. No algorithm achieves $o(T)$ regret against the optimal thresholding policy, which scores 1 every round [9, Lemma 3.1].

Given any algorithm ALG for Problem 18, run it on the instance above with $z_t = \omega_t$, take $g_t = p_{t,1}$ as the thresholding guess, and set $a_t = \alpha_1$ if $y_t = +1$ and $a_t = \alpha_2$ if $y_t = -1$. By Equation (29) the per-round payoff of ALG equals $u^{\text{OLT}}(\omega_t, g_t)$.

The benchmark. The label y_t is a deterministic function of $z_t = \omega_t$, so all rounds sharing a context share an attacker type; hence the optimal policy is $\pi^*(z) = (1, 0)$ for $z > s$ and $\pi^*(z) = (0, 1)$ for $z \leq s$, which scores $U_d = 1$ in every round. Therefore

$$R(T) = T - \sum_{t=1}^T u^{\text{OLT}}(\omega_t, g_t) = R^{\text{OLT}}(T).$$

If ALG had $o(T)$ contextual Stackelberg regret we would obtain an $o(T)$ -regret online linear thresholding algorithm, contradicting Harris et al. [9, Lemma 3.1]. Nothing in the argument uses ignorance of K , so the conclusion holds even for a defender given K in advance. $\square$

Corollary 22 (The cardinality bound is powerless with side information). *For every $K_{\max} \geq 2$ there is a contextual repeated SSG with $|K| \leq K_{\max}$ on which no algorithm is no-regret. In particular, the device that makes Problem 4 tractable — replacing “ K unknown” by “ K unknown but $|K| \leq K_{\max}$ ” — does not make Problem 18 tractable.*

It is worth being explicit about where the hardness sits, because it is not where one might expect. The instance of Theorem 21 has *context-independent* payoffs for both players: the game played at every round is literally the same 2×2 game. What the context does is index the benchmark. Competing with the best *policy* $\pi^* : \mathcal{Z} \rightarrow \mathcal{P}$ rather than with the best fixed strategy is what makes the problem hard, because the adversary can encode the identity of a_t inside z_t through a relationship that is itself not learnable online. Bounding $|K|$ limits how many distinct attacker behaviours exist; it says nothing about how they are correlated with the context, and the correlation is the difficulty. Following Harris et al. [9], we therefore restrict the adversary; Table 2 summarises the resulting picture, which Section 9.4 discusses.

Setting	Known K	Unknown K , $ K \leq K_{\max}$
No contexts	$O(\sqrt{T(n^2 + k \log n)})$ [2, Thm. 5.1]	$O(\sqrt{T K_{\max} \Lambda})$ Theorem 6
Adv. contexts, adv. types	$\Omega(T)$ [9, Thm. 3.2]	$\Omega(T)$ even if $K_{\max} = 2$ (Theorem 21)
Adv. contexts, stoch. types	$O(\min\{K, n\}\sqrt{T \log T})$ [9, Cor. 4.7–4.8]	$O(n\sqrt{T \log T})$ Theorem 26 — <i>no cost</i>
Stoch. contexts, adv. types	$O(\sqrt{KT \log T} + K)$ [9, Thm. 4.11]	$O(K_{\max}\sqrt{T \log T} + K_{\max}^2)$ Theorem 23 — <i>cost</i> $\sqrt{K_{\max}}$

Table 2: Regret rates for repeated Stackelberg security games with and without side information, for a known and an unknown attacker-type set. The asymmetry in the last two rows is explained in Section 9.3.5: an adversarial type sequence forces the learner to restart when its catalogue grows, whereas a stochastic one lets a single cumulative estimator absorb the growth.

Algorithm 3 CONTEXTUAL-GROW-CATALOGUE (stochastic contexts).

Require: horizon T , game primitives, $\delta \leftarrow 1/T$

- 1: $C \leftarrow \emptyset$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: observe z_t and compute $\mathcal{E}_{z_t}(C; \delta)$
- 4: sample $\pi_t \sim p_t$ from the current Hedge distribution over $\Pi(C)$; play $\mathbf{p}_t = \pi_t(z_t)$
- 5: observe a_t ; if $a_t \notin C$, record its dossier
- 6: for each $\pi^{(\omega)} \in \Pi(C)$ set $\ell_t[\pi^{(\omega)}] \leftarrow -U_d(b_{a_t}(z_t, \pi^{(\omega)}(z_t)), \pi^{(\omega)}(z_t); z_t)$ and update Hedge
- 7: **if** $a_t \notin C$ **then** $\triangleright$ catalogue growth: rebuild and restart
- 8: $C \leftarrow C \cup \{a_t\}$; $\Pi(C) \leftarrow \{\pi^{(\omega)}\}_{\omega \in \Omega(C)}$; reset Hedge on $\Pi(C)$
- 9: **end if**
- 10: **end for**

9.3.4 Stochastic contexts, adversarial types

Here $z_1, \dots, z_T$ are drawn i.i.d. from an unknown distribution $\mathcal{D}_z$ on $\mathcal{Z}$, while $a_1, \dots, a_T \in K^T$ is chosen in advance by an oblivious adversary that knows $\mathcal{D}_z$ (the same adversary model as Theorem 6). Performance is measured by $\mathbb{E}[R(T)]$, the expectation being over the draw of the contexts and the algorithm's randomisation, against the policy π^* that is optimal given $\mathcal{D}_z$ and $a_1, \dots, a_T$ [9, Definition 4.9].

Harris et al. [9, Theorem 4.11] solve the known- K case by running Hedge over a finite policy class indexed by a grid of weight vectors on the simplex Δ_K . When K is unknown, that class is not available at the start; the natural fix is the one this paper already uses — grow it, and restart.

Concretely, for a catalogue C let

$$\Omega(C) = \{\omega \in \Delta_{|C|} : T\omega[\alpha] \in \mathbb{N} \forall \alpha \in C\},$$

$$\pi^{(\omega)}(z) = \arg \max_{\mathbf{p} \in \mathcal{E}_z(C; \delta)} \sum_{\alpha \in C} \omega[\alpha] U_d(b_\alpha(z, \mathbf{p}), \mathbf{p}; z),$$

and $\Pi(C) = \{\pi^{(\omega)} : \omega \in \Omega(C)\}$, so that

$$|\Pi(C)| \leq |\Omega(C)| = \binom{T + |C| - 1}{|C| - 1} \leq T^{|C|} \leq T^{K_{\max}}. \quad (30)$$

Theorem 23 (Stochastic contexts, adversarial types, unknown K). *Let the defender play Algorithm 3 in the setting of Problem 18 with i.i.d. contexts and an oblivious adversarial type sequence with $|K| \leq K_{\max}$. Then*

$$\mathbb{E}[R(T)] = O\left(K_{\max}\sqrt{T \log T} + K_{\max}^2\right).$$

Proof. Because the adversary is oblivious, the discovery rounds $\tau_1 < \dots < \tau_{K'}$, the epochs $\text{Ep}_1, \dots, \text{Ep}_{K'}$ of Equation (8), their catalogues C_i and their lengths L_i are all deterministic, and $K' \leq K_{\max}$. Fix an epoch i ; by construction $a_t \in C_i$ for every $t \in \text{Ep}_i$.

(a) *From π^* to a policy into $\mathcal{E}(C_i; \delta)$.* Harris et al. [9, Lemma 4.4], applied at each context with the catalogue C_i and the sub-sequence $(a_t)_{t \in \text{Ep}_i}$, produces a policy $\pi_i^{(E)}$ with $\pi_i^{(E)}(z) \in \mathcal{E}_z(C_i; \delta)$ and

$$\sum_{t \in \text{Ep}_i} \left(U_d(b_{a_t}(z_t, \pi^*(z_t)), \pi^*(z_t); z_t) - U_d(b_{a_t}(z_t, \pi_i^{(E)}(z_t)), \pi_i^{(E)}(z_t); z_t) \right) \leq \delta L_i.$$

The hypothesis of that lemma is exactly that only types of C_i occur in the sub-sequence, which is Equation (9). Summing over epochs, the total cost is $\delta \sum_i L_i \leq \delta T = 1$.

(b) *From $\pi_i^{(E)}$ to the grid $\Pi(C_i)$.* Since the contexts are i.i.d. and the type sequence is fixed, for any policy π

$$\mathbb{E}_z \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(z_t, \pi(z_t)), \pi(z_t); z_t) \right] = \sum_{\alpha \in C_i} \nu_\alpha \mathbb{E}_{z \sim \mathcal{D}_z} \left[U_d(b_\alpha(z, \pi(z)), \pi(z); z) \right], \quad (31)$$

where $\nu_\alpha = |\{t \in \text{Ep}_i : a_t = \alpha\}|$ and $\sum_\alpha \nu_\alpha = L_i$; this is Harris et al. [9, Lemma 4.10] specialised to a fixed type sequence. So the epoch value of a policy depends on it only through the $|C_i|$ numbers $v_\alpha(\pi) = \mathbb{E}_z[U_d(b_\alpha(z, \pi(z)), \pi(z); z)] \in [0, 1]$, weighted by ν . Let $\omega^* \in \Omega(C_i)$ round the probability vector ν/L_i onto the grid by setting $T\omega^*[\alpha] = \lfloor T\nu_\alpha/L_i \rfloor$ and distributing the at most $|C_i|$ leftover units arbitrarily, so that $\sum_\alpha |\omega^*[\alpha] - \nu_\alpha/L_i| \leq |C_i|/T$. The policy $\pi^{(\omega^*)}$ maximises the ω^* -weighted objective over $\mathcal{E}_z(C_i; \delta)$ pointwise in z , hence over the class containing $\pi_i^{(E)}$; transferring from the ω^* -weighting to the ν -weighting costs at most $2L_i \cdot |C_i|/T \leq 2|C_i| \leq 2K_{\max}$ because $v_\alpha(\cdot) \in [0, 1]$. Therefore

$$\mathbb{E}_z \left[\sum_{t \in \text{Ep}_i} U_d(\cdot, \pi^{(\omega^*)}) \right] \geq \mathbb{E}_z \left[\sum_{t \in \text{Ep}_i} U_d(\cdot, \pi_i^{(E)}) \right] - 2K_{\max}.$$

(c) *Hedge inside the epoch.* Algorithm 3 runs Hedge over the fixed class $\Pi(C_i)$ for L_i rounds with losses in $[-1, 0]$, so

Proposition 3 and Equation (30) give

$$\begin{aligned} \max_{\pi \in \Pi(C_i)} \sum_{t \in \text{Ep}_i} U_d(\cdot, \pi) - \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(\cdot, \mathbf{p}_t) \right] \\ \leq c_1 \sqrt{L_i \log |\Pi(C_i)|} \leq c_1 \sqrt{L_i K_{\max} \log T}. \end{aligned}$$

(d) *Assembling.* Adding (a)–(c) over the $K' \leq K_{\max}$ epochs, and paying at most 2 per discovery round exactly as in Step 6 of Section 6,

$$\begin{aligned} \mathbb{E}[R(T)] &\leq 1 + 2K_{\max} + 2K_{\max}^2 \\ &\quad + c_1 \sqrt{K_{\max} \log T} \sum_{i=1}^{K'} \sqrt{L_i}. \end{aligned}$$

Cauchy–Schwarz gives $\sum_i \sqrt{L_i} \leq \sqrt{K'T} \leq \sqrt{K_{\max}T}$, so the last term is $O(K_{\max} \sqrt{T \log T})$, and the claim follows. $\square$

Compared with the known- K rate $O(\sqrt{KT \log T} + K)$ of Harris et al. [9, Theorem 4.11], Theorem 23 pays a multiplicative $\sqrt{K_{\max}}$ and an additive $K_{\max}^2$. The multiplicative factor arises in exactly the same place as in Theorem 6 — the Cauchy–Schwarz step over the epochs forced by restarting Hedge — so the price of an unknown catalogue is the same in the contextual and non-contextual worlds.

Remark 24 (Adaptive type adversaries). Theorem 23 is stated for an oblivious type adversary, matching Theorem 6. It extends to an adaptive adversary that chooses a_t knowing $\mathcal{D}_z$ and $z_1, \dots, z_{t-1}$ but not z_t , at the cost of one extra observation: both a_t and the epoch index of round t are measurable with respect to $\sigma(a_1, \dots, a_t)$, and the adversary sees only $z_1, \dots, z_{t-1}$ when it picks a_t ; hence z_t is independent of the pair $(\mathbf{1}\{a_t = \alpha\}, \mathbf{1}\{t \in \text{Ep}_i\})$ and Equation (31) survives with ν_α replaced by $\sum_t \Pr[a_t = \alpha, t \in \text{Ep}_i]$. The rounding argument in step (b) is then applied to those expectations, and the bound is unchanged.

9.3.5 Stochastic types, adversarial contexts

Now reverse the relaxation: $a_1, \dots, a_T$ are drawn i.i.d. from an unknown distribution $\mathcal{F}$ supported on the unknown set K , while the contexts are adversarial (possibly adaptively, knowing $\mathcal{F}$ and $a_1, \dots, a_{t-1}$ but not a_t). Write

$$q(z, \mathbf{p})[i] = \Pr_{a \sim \mathcal{F}}[b_a(z, \mathbf{p}) = i], \quad i \in N,$$

for the induced distribution over attacked targets, and let

$$\hat{q}_t(z, \mathbf{p})[i] = \frac{1}{t-1} \sum_{\tau < t} \mathbf{1}\{b_{a_\tau}(z, \mathbf{p}) = i\}$$

be its empirical estimate. The defender plays greedily,

$$\mathbf{p}_t \in \arg \max_{\mathbf{p} \in \mathcal{E}_z(C_t; \delta)} \hat{V}_t(z_t, \mathbf{p}), \quad (32)$$

where $\hat{V}_t(z, \mathbf{p}) = \sum_{i \in N} U_d(i, \mathbf{p}; z) \hat{q}_t(z, \mathbf{p})[i]$ and $\delta = 1/T$. Note that $\hat{q}_t$ is computable: it only ever evaluates b_{a_τ} for types already observed, whose dossiers the defender holds.

The reason this regime behaves completely differently from Section 9.3.4 is the following elementary observation, which says that the defender’s *estimated* objective is piecewise linear with respect to precisely the partition the defender can compute.

Lemma 25 (Greedy over the catalogue is δ -optimal for the estimate). *Fix t and z , and let $C_t = \{a_1, \dots, a_{t-1}\}$. Then $\hat{V}_t(z, \cdot)$ is constant in its $\hat{q}_t$ factor on every cell $\mathcal{P}_\sigma(z; C_t)$ and therefore linear in $\mathbf{p}$ on that cell. Consequently*

$$\max_{\mathbf{p} \in \mathcal{E}_z(C_t; \delta)} \hat{V}_t(z, \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \hat{V}_t(z, \mathbf{p}) - \delta.$$

Proof. For $\tau < t$ we have $a_\tau \in C_t$, so $b_{a_\tau}(z, \mathbf{p}) = \sigma(a_\tau)$ for every $\mathbf{p} \in \mathcal{P}_\sigma(z; C_t)$; hence $\hat{q}_t(z, \cdot)$ is constant on the cell, and $\hat{V}_t(z, \cdot)$ is a fixed convex combination of the linear functions $U_d(i, \cdot; z)$ there. Let $\mathbf{p} \in \mathcal{P}$ be arbitrary and σ its cell. Maximising the linear function $\hat{V}_t(z, \cdot)$ over the closed convex polytope $\text{cl}(\mathcal{P}_\sigma(z; C_t))$ is attained at an extreme point $\mathbf{p}'$, and $\mathcal{E}_z(C_t; \delta)$ contains a point $\mathbf{p}'' \in \mathcal{P}_\sigma(z; C_t)$ with $\|\mathbf{p}' - \mathbf{p}''\|_1 \leq \delta$. Since $u_d^c, u_d^u \in [0, 1]$ we have $|U_d(i, \mathbf{p}'; z) - U_d(i, \mathbf{p}''; z)| \leq \|\mathbf{p}' - \mathbf{p}''\|_1 \leq \delta$ for every i , so $\hat{V}_t(z, \mathbf{p}'') \geq \hat{V}_t(z, \mathbf{p}') - \delta \geq \hat{V}_t(z, \mathbf{p}) - \delta$. $\square$

Lemma 25 is the crux. It does *not* compare the defender’s feasible set with the finer set $\mathcal{E}_z(K; \delta)$ that a defender knowing K could use. It says something better: relative to the objective the defender is actually optimising, the coarse set $\mathcal{E}_z(C_t; \delta)$ is already optimal up to δ , because the estimator and the partition are built from the same observed types and refine together. There is therefore no need to restart anything when a new type arrives, and no epoch decomposition is required.

Theorem 26 (Stochastic types, adversarial contexts, unknown K). *Let the defender play Equation (32) in the setting of Problem 18 with i.i.d. attacker types and adversarial contexts, and $|K| \leq K_{\max}$. Then*

$$\mathbb{E}[R(T)] \leq 1 + 2 \sum_{t=1}^T \mathbb{E}[\text{TV}(q_t^*, \hat{q}_t^*) + \text{TV}(q_t, \hat{q}_t)],$$

where $q_t^* = q(z_t, \pi^*(z_t))$, $\hat{q}_t^* = \hat{q}_t(z_t, \pi^*(z_t))$, $q_t = q(z_t, \mathbf{p}_t)$ and $\hat{q}_t = \hat{q}_t(z_t, \mathbf{p}_t)$. Hence, with the concentration accounting of Harris et al. [9, Corollary 4.8],

$$\mathbb{E}[R(T)] = O(n \sqrt{T \log T}).$$

The bound contains no dependence on $K_{\max}$ whatsoever.

Proof. Fix t and abbreviate $z = z_t$. Write $\mathbb{E}_t[\cdot]$ for expectation over $a_t \sim \mathcal{F}$ given the history and $\hat{\mathbb{E}}_t[\cdot]$ for the same expression with q replaced by $\hat{q}_t$, so that $\hat{\mathbb{E}}_t[U_d(b_{a_t}(z, \mathbf{p}), \mathbf{p}; z)] = \hat{V}_t(z, \mathbf{p})$. Since $U_d \in [0, 1]$, for any $\mathbf{p}$

$$|\mathbb{E}_t[U_d(b_{a_t}(z, \mathbf{p}), \mathbf{p}; z)] - \hat{V}_t(z, \mathbf{p})| \leq 2 \text{TV}(q(z, \mathbf{p}), \hat{q}_t(z, \mathbf{p})). \quad (33)$$

By Lemma 25 applied with the comparator $\pi^*(z) \in \mathcal{P}$, $\hat{V}_t(z, \mathbf{p}_t) \geq \hat{V}_t(z, \pi^*(z)) - \delta$. Combining this with Equation (33) at $\mathbf{p} = \pi^*(z)$ and at $\mathbf{p} = \mathbf{p}_t$,

$$\begin{aligned} \mathbb{E}_t[U_d(b_{a_t}(z, \pi^*(z)), \pi^*(z); z)] &\leq \mathbb{E}_t[U_d(b_{a_t}(z, \mathbf{p}_t), \mathbf{p}_t; z)] \\ &\quad + \delta + 2 \text{TV}(q(z, \pi^*(z)), \hat{q}_t) + 2 \text{TV}(q(z, \mathbf{p}_t), \hat{q}_t). \end{aligned}$$

Summing over t , taking expectations and using $\delta T = 1$ gives the first display. For the second, Harris et al. [9, Corollary 4.8] show by Hoeffding’s inequality that $\mathbb{E}[\text{TV}(q(z, \mathbf{p}), \hat{q}_t(z, \mathbf{p}))] \leq n \sqrt{\log(2T)/(t-1) + 1/(2T)}$, and $\sum_{t=2}^T (t-1)^{-1/2} = O(\sqrt{T})$. $\square$

Remark 27 (On the uniformity of the concentration). The bound $\mathbb{E}[\text{TV}] \lesssim n\sqrt{\log T/t}$ quoted above is the one stated by Harris et al. [9, Corollary 4.8]; their union bound is taken over rounds and targets. A bound that is uniform over all pairs $(z, \mathbf{p})$ needs a union over the distinct values the estimator can take, i.e. over the at most $n^{|C_t|} \leq n^{K_{\max}}$ best-response profiles of Definition 19, which replaces $\log T$ by $\log T + K_{\max} \log n$ and yields the self-contained bound $\mathbb{E}[R(T)] = O(n\sqrt{T(\log T + K_{\max} \log n)})$. Either way the point stands: the term is present verbatim in the known- K analysis, so not knowing K adds nothing.

If instead the defender estimates the type distribution directly — the analogue of Harris et al. [9, Corollary 4.7], with $\hat{p}_t(\alpha) = \frac{1}{t-1} \sum_{\tau < t} \mathbf{1}\{a_\tau = \alpha\}$ supported on C_t — then unknown K does cost something, but only an additive constant per type.

Proposition 28 (Type-frequency estimator). *With the estimator $\hat{p}_t$ above, which assigns zero mass to every type not yet observed,*

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[\text{mass of } \mathcal{F} \text{ outside } C_t] \\ = \sum_{\alpha \in K} \sum_{t=1}^T \mathcal{F}(\alpha) (1 - \mathcal{F}(\alpha))^{t-1} \leq |K| \leq K_{\max}, \end{aligned}$$

and consequently the greedy algorithm driven by $\hat{p}_t$ has $\mathbb{E}[R(T)] = O(K_{\max} \sqrt{T \log T} + K_{\max})$.

Proof. Type α is absent from C_t exactly when it has not occurred in the first $t-1$ rounds, which has probability $(1 - \mathcal{F}(\alpha))^{t-1}$; the expected missing mass at round t is $\sum_{\alpha} \mathcal{F}(\alpha) (1 - \mathcal{F}(\alpha))^{t-1}$. Summing the geometric series over t gives at most 1 per type. The missing mass bounds the extra total variation incurred by $\hat{p}_t$ relative to the known-support estimator, so the regret bound of Harris et al. [9, Corollary 4.7] picks up this additive term. $\square$

Proposition 28 makes the mechanism explicit: a type that is rare enough to stay undiscovered is, for exactly that reason, rare enough to be harmless. This is why stochastic attackers forgive an unknown catalogue and adversarial ones do not.

9.4 Summary and open problems

Table 2 collects the picture for Section 9.3. The non-contextual row is included for reference; $\Lambda = n^2 + K_{\max} \log n$ as in Equation (20).

Three questions are left open.

Bandit feedback with an unknown catalogue. Both Balcan et al. [2, Sec. 6] and Harris et al. [9, Sec. 5] pass from full to bandit feedback by constructing a barycentric spanner [1] for the space of best-response indicator vectors, which lives in $\mathbb{R}^K$ and therefore has a basis of size K . With an unknown catalogue that space *grows*, so the spanner must be recomputed at every discovery round, and one has to bound the loss from discarding the estimates built on the old basis. Combined with the change-point detection needed to notice a new type at all, we expect a bound of the form $O(T^{2/3} \text{poly}(n, K_{\max}))$ plus a detection-delay term.

Lower bounds. Theorem 23 and Theorem 6 both pay $\sqrt{K_{\max}}$ relative to their known- K counterparts, and we do not know whether that factor is necessary. Likewise, Theorem 16 leaves open whether the blind-round cost $\Theta(K_{\max} P)$ of Theorem 13 is real when $K_{\max}$ is small (Remark 17).

Combining the two directions. Problem 11 and Problem 18 are orthogonal relaxations and can be imposed together: periodic revelation with side information. Theorem 26 suggests this is benign in the stochastic-attacker regime — the greedy estimator never needed the catalogue to be current — whereas Theorem 23 would inherit both the $\sqrt{K_{\max}}$ restart factor and the blind-round term $K_{\max} P$.

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